Problem Statement

AI Agent for Smart Farming Advice

Introduction

Agriculture is one of the most important sectors contributing to food production and economic development. However, many small-scale farmers face challenges in accessing reliable and timely agricultural information. Decisions related to crop selection, soil management, fertilizer usage, pest control, and farming practices often require expert knowledge, which may not always be available to farmers.

Problem Statement

Farmers frequently encounter difficulties in obtaining accurate agricultural guidance required for effective decision-making. Limited access to agricultural experts, lack of awareness about modern farming techniques, and difficulties in understanding technical information can negatively impact crop productivity and farm income. Additionally, information related to crop cultivation, soil health, fertilizers, and pest management is often scattered across multiple sources, making it difficult for farmers to access and utilize it efficiently.

Proposed Solution

The proposed solution is an AI-powered Smart Farming Advisor developed using IBM watsonx Orchestrate and a Retrieval-Augmented Generation (RAG) knowledge system. The system retrieves relevant information from agricultural documents and provides farmers with accurate, knowledge-based recommendations. Farmers can interact with the advisor through a simple chat interface and receive guidance on crop selection, soil management, fertilizer application, and pest control.

Objectives

- To provide reliable agricultural guidance to farmers.
- To recommend suitable crops based on farming conditions.
- To assist in soil health and fertilizer management.
- To support pest and disease identification and control.
- To retrieve accurate information from agricultural knowledge sources.
- To promote sustainable and smart farming practices.

Expected Outcomes

The Smart Farming Advisor will help farmers make informed agricultural decisions, improve productivity, reduce risks, and enhance the adoption of modern farming practices. By providing easy access to agricultural knowledge, the system aims to bridge the information gap and support sustainable agricultural development.